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Unlocking U.S. Renewable Potential: A Deeper Understanding Through an Integrated Techno-Economic and Life-Cycle Assessment of Subsurface Hydrogen and Synthetic Geothermal Storage

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Abstract

Renewable energy is gaining attention due to the need to lower carbon emissions. However, the fluctuating nature of wind and solar energy strains the electric grid, leading to significant energy losses in several regions in the United States. Although energy supply can be managed through storage technologies, the lack of efficient methods and understanding of their impact has limited their integration into the electric grid.

This study explores the potential of storing curtailed renewable energy in major U.S. electric regions: ERCOT, CAISO, MISO, SPP, NYISO, ISO-NE, and PJM, through an integrated techno-economic and life-cycle assessment of two emerging storage technologies: subsurface hydrogen (H2) and synthetic geothermal storage. Using the yearly average of curtailed energy from 2007 to 2023, the total efficiencies, levelized costs and greenhouse gas emissions can be quantified, offering a comprehensive evaluation of their feasibility. Various phases were analyzed including H2 and synthetic geothermal fluid production, processing, storage, and power production. Reservoir simulation model was also utilized to determine the withdrawal efficiencies.

The results of this study highlight the potential recovery percentages for subsurface H2 storage and synthetic geothermal storage across major U.S. regions, providing a detailed analysis of their efficiency at varying depths. The study also presents the expected minimal levelized cost, measured in $/MWh, offering a clear assessment of the technical and economic feasibility of each storage option. A comparative evaluation across regions can reveal the extent to which renewable energy could have been stored. Additionally, the anticipated greenhouse gas emissions are assessed, allowing for a thorough comparison of the environmental impacts in different regions, thus providing a holistic view of the technologies' potential benefits and challenges.

This study offers a comparative exploration of the potential for integrating subsurface H2 and synthetic geothermal storage into the U.S. electric grid, evaluating their feasibility from technical, economic, and environmental perspectives across different regions.

Introduction

Electrical power generation from renewable sources such as wind and solar, is a key strategy in reducing greenhouse gas emissions and dependence on fossil fuels. However, fluctuations in their supply poses challenges to grid stability. As renewable capacity continues to expand, so does energy curtailment—where excess power cannot be delivered or used due to grid limitations. Integrating long-duration energy storage into the power system is essential to balance supply and demand, minimize curtailment, reduce emissions, and maximize the utilization of renewable energy.

In 2020, the United States ranked second globally in renewable energy capacity, reaching 292 GW, following China's 895 GW (Niaz et al., 2022). Over the past decade, the U.S. has experienced substantial growth in wind and solar energy production, accompanied by a rise in curtailments. Curtailment levels vary across the country and are reported by regional Independent System Operators (ISOs), including the Electric Reliability Council of Texas (ERCOT), California ISO (CAISO), Southwest Power Pool (SPP), Midcontinent Independent System Operator (MISO), and New York Independent System Operator (NYISO), ISO New England (ISO-NE) and PJM Interconnection (formerly for Pennsylvania, New Jersey, and Maryland but has expanded to include parts or all of 13 states and the District of Columbia). Addressing regional curtailment concerns requires efficient storage solutions.

Among the emerging options for long-duration energy storage are subsurface H_2 storage and synthetic geothermal storage. Excess renewable energy can be converted into "green hydrogen" via electrolysis, a process that generates little to no direct greenhouse gas emissions. This hydrogen can then be stored in suitable subsurface geological formations—such as depleted hydrocarbon reservoirs or saline aquifers—offering large-scale, long-term storage capacity. Optimal storage formations typically include dipping or anticline-shaped reservoirs that are well-sealed both vertically and laterally, which helps retain the hydrogen and minimize water production during recovery (Okoroafor, et al. 2022). Hydrogen's versatility further enhances its value, as it can be used for electricity generation, industrial processes, heating, transportation, and the production of synthetic fuels such as ammonia.

Synthetic geothermal storage offers another pathway to store curtailed energy, particularly from solar sources. Solar radiation can be captured as either electricity through photovoltaic (PV) panels or as thermal energy via solar thermal systems. PV-generated electricity can be used to power electric boilers that produce steam, which is then stored for later power generation. Alternatively, solar thermal systems directly heat fluids such as water or air, which can be stored in underground porous media. In both approaches, thermal energy is injected and stored in geologic formations for later recovery.

Selecting an optimal energy storage method depends on several factors, including cost, environmental impact, storage scale (small or large), and charge/discharge duration (Argyrou et al., 2016). Storage durations can range from a few hours (short-term) to several weeks or months (long-term), depending on the application and technology.

This study focuses on the technical, economic, and environmental feasibility of underground H2 storage and synthetic geothermal storage in porous media as complementary solutions to renewable energy integration across various U.S. ISO regions. By incorporating a techno-economic and life cycle assessment (LCA) framework, this work aims to evaluate the potential of these subsurface storage technologies to reduce renewable energy curtailment and support a sustainable low-carbon energy future.

Methodology

Wind and solar curtailment for each U.S. ISO region were estimated using data from Wiser et al. (2024), Seel et al. (2024), and the U.S. Energy Information Administration (EIA, 2024). Wiser et al. (2024) reported annual wind curtailment percentages for ISO regions from 2007 to 2023, while Seel et al. (2024) provided solar curtailment percentages from 2015 to 2023. The EIA (2024) supplied electricity generation data (in MWh) by state and energy source from 1990 to 2023. To estimate electricity generation for each ISO region, generation values from all states within each region were aggregated. The corresponding curtailment percentages were then applied to calculate wind and solar curtailment from 2007 to 2023, in MWh, for each ISO region: Electric Reliability Council of Texas (ERCOT), California ISO (CAISO), Southwest Power Pool (SPP), Midcontinent Independent System Operator (MISO), and New York Independent System Operator (NYISO), ISO New England (ISO-NE) and PJM Interconnection (formerly for Pennsylvania, New Jersey, and Maryland but has expanded to include parts or all of 13 states and the District of Columbia).

The calculated curtailment data were used to evaluate the potential for storing excess wind and solar energy in each ISO region through subsurface H_2 and synthetic geothermal storage. This assessment considered technical, economic, and environmental factors, following a methodology similar to that of Tayyib et al. (2025). The primary objective of the technical evaluation was to quantify key parameters across the various phases of each storage system to determine the overall system efficiency. For subsurface hydrogen storage, the analysis covered several major phases, and associated parameters as summarized in Table 1, including:

1. **Production:** H_2 generation using alkaline electrolyzer and proton exchange membrane (PEM) electrolyzer, along with their respective water requirements.
2. **Processing:** Electrical energy consumption for hydrogen compression and pumping.
3. **Storage:** Injection into shallow aquifers (500 m depth at 50 bar) and deep aquifers (2000 m depth at 200 bar), including consideration of withdrawal volumes.
4. **Power Generation:** Conversion of stored hydrogen back into electricity using a fuel cell.

Table 1—Summary of the parameters quantified, and corresponding equations used in the technical assessment of subsurface H# storage technology, adapted from Tayyib et al. (2025).

Parameter	Equations
Amount of H_2 produced [KgH_2]	$m_{H2\ Produced} = \text{Curtailed Energy} \div h_{electrolyser}$
Amount of water required for electrolysis [KgH_2O]	$\text{Water Requirement} = m_{H2\ produced} \times WC$
Energy required for H_2 compression [kWh]	$\text{Energy Required for Compression}\left(\frac{E}{Qin \times t \times \rho_{H2}}\right) = \left(\frac{2x10-5}{\gamma \times \rho_{H2}} \times \left[\left(\frac{P_{in}}{P_{out}}\right)^{\gamma} - 1\right]\right) \times \frac{277\ kWh}{GJ}$
Energy required for H_2 pumping [kWh]	$\text{Energy Required for Pumping (P)} = \eta_p \rho V_{H2} gh \times \frac{1\ N}{kg \cdot \frac{m}{S^2}} \times \frac{W}{\frac{mN}{s}} \times \frac{1\ MW}{10^6\ W} \times 24\ hr$
Injection volume [m^3/d]	$\text{Injection Volume} = m_{H2produced} \div \rho_{H2}$ at specific pressure & temperature $\text{Temperature at a specific depth} = T_{surf} + \frac{\Delta T}{\Delta D} \times D$
Withdrawal volume [m^3/d]	$\text{Withdrawal Volume} = \text{Injection Volume} \times \text{Withdrawal Effeciency}$ $\text{Withdrawal Mass} = \text{Withdrawal Volume} \times \rho_{H2sub}$
generated [MWh]	$\textit{Power Generated} = \text{Withdrawal mass} \times LHV$
System's Total Efficiency [%]	$h_{system} = \frac{\text{Power Generated}}{(\text{Curtailed Energy} + \text{Electrical Energy Consumption})} \times 100\%$

Similarly, for synthetic geothermal storage, the analysis encompassed several key phases, and associated parameters as summarized in Table 2, including:

1. **Thermal Energy Production:** Heat generation at 200°C using either an electric-powered hot water/steam boiler or concentrated solar power (CSP) technology.
2. **Processing:**
   - For CSP: steam condensation and water pumping.
   - For electric boilers: water heating, pressurization, and pumping.
3. **Storage:** Injection into shallow reservoirs (500 m depth at 50 bar), and deep reservoirs (2000 m depth at 200 bar), including consideration of withdrawal volumes.
4. **Power Generation:** Electricity production using a turbine coupled with a generator.

Table 2—Summary of the parameters quantified, and the equations used for the technical assessment for synthetic goethermal storage technology.

Parameter	Equations
Mass of water that can be heated [Kg]	$Q = C_v\, m_{water}\, \Delta T$ $Q_{total} = Q_1 + Q_2 + Q_2$ $m_{water} = \dfrac{Qtotal}{C_{p1}\,\Delta T1 + H + C_{p2}\Delta T2}$
Injection volume [m3/d]	$\text{Injection Volume } (V_{H2O}) = \dfrac{m_{water}}{month} \times \dfrac{1 \text{ month}}{\text{number of Days}} \div \rho_{water}$
Energy consumption for water pumping [MWh]	$\text{Energy Required for Pumping (P)} = \eta_p \rho_{water} V_{H2O} gh \times \dfrac{1\text{ N}}{\text{kg}\cdot\frac{\text{m}}{\text{S}^2}} \times \dfrac{\text{W}}{\frac{\text{mN}}{\text{s}}} \times \dfrac{1\text{ MW}}{10^6\text{ W}} \times 24\, hr$
Energy consumption for heating, (for electrical boiler) [MWh]	$\text{Energy Required for Heating} = m_{water} \times Req_{Heat}$
Energy consumption for condensation (for CSP) [MWh]	$\text{Energy Required for Condensation} = m_{water} \times Q_{cond}$
Reservoir simulation for withdrawal volume [m3/d]	$\text{Withdrawal Efficiency} = \dfrac{Q_{out}}{Q_{in}} = \dfrac{m_{out} \times (T_{out} - T_{surf})}{m_{in} \times (T_{in} - T_{surf})}$ $\text{Withdrawal Volume} = \text{Injection Volume} \times \text{Withdrawal Efficiency}$
The power generated [MWh]	$\text{Withdrawal Mass [Kg]} = \text{Withdrawal Volume} \left[\frac{m^3}{d}\right] \times \frac{\text{No. of Days}}{1\text{ Month}} \times \rho_{water}$ $\text{Power Produced} = \text{Withdrawal Mass [Kg]} \times H \times \eta_{turb} \times \eta_{gen}$
System's Total Efficiency [%]	$h_{system} = \dfrac{\text{Power Generated}}{(\text{Curtailed Energy} + \text{Electrical Energy Consumption})} \times 100\%$

For the storage phase, withdrawal efficiencies were estimated using a reservoir simulation conducted in ECLIPSE E300, following a methodology similar to that of Okoroafor et al. (2022). The simulation grid comprised 150,000 cells, arranged as 50 × 50 cells in the x and y directions and 60 cells in the z direction. Each cell represented a volume of 3,200 m³ (40 m × 40 m × 2 m). A single well, located near the center of the model domain, was used for both injection and production. Table 3 summarizes the reservoir properties applied in the simulation.

Table 3—Summary of the parameters used for reservoir simulation to determine withdrawal efficiencies.

Parameters	Value	Units
Width and length of the reservoir	2000	m
Thickness of the reservoir	120	m
Porosity of the reservoir	25	%
Horizontal reservoir permeability	300	mD
Vertical reservoir permeability	30	mD

Two scenarios were simulated involving hot water injection into shallow and deep reservoirs. Key parameters such as the lowest temperature drop (T_{out}) during production, surface temperature (T_{surf}), and the volumes of water injected (m_{in}) and produced (m_{out}), were used to calculate the withdrawal efficiencies for the synthetic geothermal storage. The maximum, minimum, and average values of these parameters were estimated and contributed to calculating the overall system efficiency.

Following the technical assessment, both economic and LCA were conducted to evaluate the feasibility of subsurface H_2 storage and synthetic geothermal storage technologies across various ISO regions. These assessments quantified the levelized costs and associated greenhouse gas (GHG) emissions using various factors sourced from the literature. For subsurface H_2 storage, economic parameters included capital and operating costs for wind turbines, alkaline and PEM electrolyzers, compressors, wells, pipelines, and fuel cells—drawing on data from El-Emam & Ozcan (2019), Khan et al. (2021), Chen et al. (2023), Lord et al. (2014), and Salmachi et al. (2022). For synthetic geothermal storage, parameters were based on capital and O&M costs for CSP and PV systems, electric heaters, pumps, wells, pipelines, turbines, generators, with sources including El-Emam & Ozcan (2019), Giap et al. (2022), Wagner (2011), Aviles et al. (2021), and Beckers & McCabe (2019). The LCA incorporated GHG emission factors from Ali et al. (2022), Schaefer (2022), Kramens et al. (2024), and Thomas et al. (2020), covering emissions from electricity generation, hydrogen production and transport, and geothermal system components.

Results and Discussions

Fig. 1 presents the aggregated wind and solar electricity generation by U.S. region from 2007 to 2023, measured in terawatt-hours (TWh). Fig. 1a shows that the leading regions for wind power generation are SPP, ERCOT, and MISO. Given their high generation levels, these regions also exhibit higher curtailment, consistent with findings from Wiser et al. (2023). In 2022, the highest wind curtailment rates were reported in SPP (9.2%), ERCOT (4.7%), and MISO (4.4%), while other regions experienced curtailment rates of 2% or lower. These elevated curtailment levels align with the favorable wind resource conditions in SPP and ERCOT, which experience average annual wind speeds exceeding 8 m/s at 100 meters above ground. In terms of solar power, Fig. 1b shows that CAISO, NON-ISO regions, PJM, and ERCOT lead in generation. However, among the seven ISOs, only CAISO and ERCOT report solar curtailment data. Both regions are located in areas receiving more than 5.0 kWh/m²/day of solar radiation—higher than the levels observed in other regions (Bolinger et al., 2023).

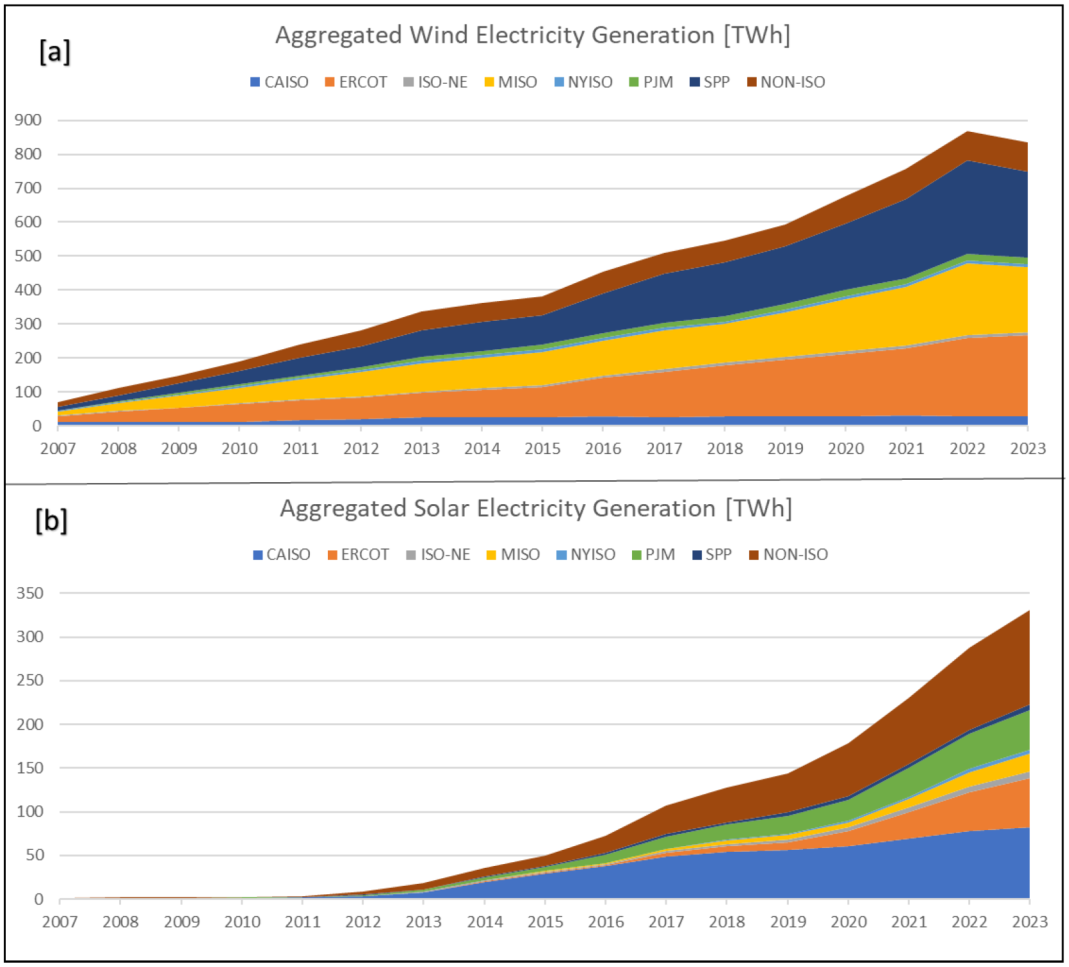

Figure 1—Aggregated U.S. electric power generation by region: (a) from wind sources; (b) from solar sources. Graphs generated using data from EIA (2024).

With the increasing trend in power generation across all U.S. regions, the estimated curtailments have also risen (Fig. 2) —an expected outcome of maintaining electric grid stability. Fig. 2a shows that the regions with the highest wind energy curtailments—SPP, ERCOT, and MISO—are also the regions with the highest wind power generation (Fig. 1a). In the case of solar energy, Fig. 2b shows that ERCOT experiences higher levels of curtailment compared to CAISO. Both figures reveal an overall increasing trend in curtailments over time. These curtailed energy volumes represent clean, renewable power that could have been utilized if efficient storage solutions were in place. As curtailments are expected to continue growing in the future (Hoza, 2022), the deployment of energy storage technologies will be critical to reducing renewable energy waste, meeting future demand, and minimizing emissions.

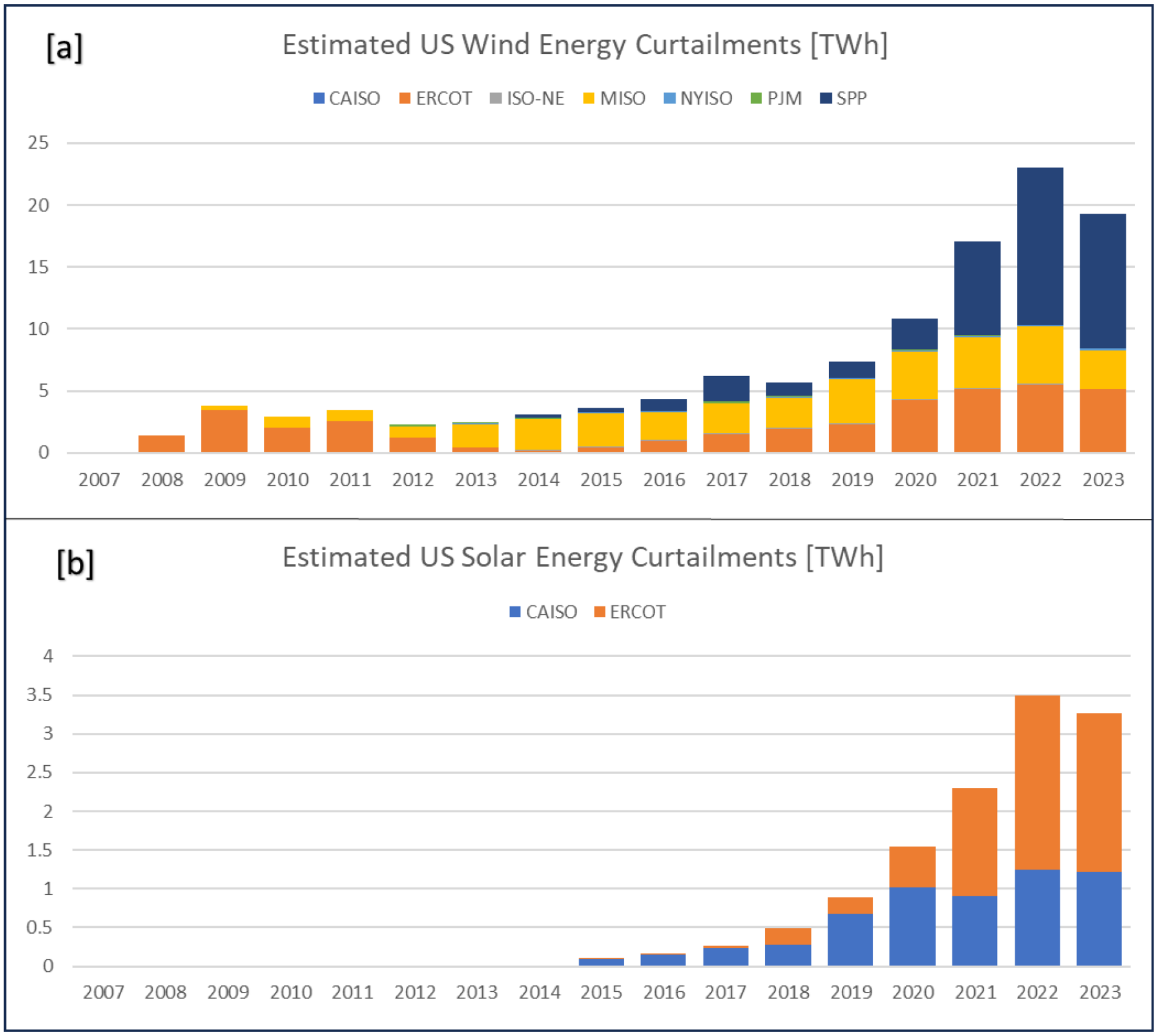

Figure 2—Estimated Average U.S. curtailments from 2007 to 2024: (a) from wind sources; (b) from solar sources. Graphs generated using data from EIA (2024), Wiser et al. (2024), and Seel et al. (2024).

Fig. 3 illustrates the regional distribution of energy in the U.S. that could have been recovered from wind and solar sources over the years. Fig. 3a shows a notable increase in SPP's contribution in recent years, accounting for approximately 56% of the total recoverable wind energy in 2023 using storage technologies, followed by ERCOT at 26% and MISO at 16%. In contrast, Fig. 3b highlights a growing share of ERCOT in recoverable solar energy, reaching about 63% in 2023, while CAISO accounts for the remaining 37%.

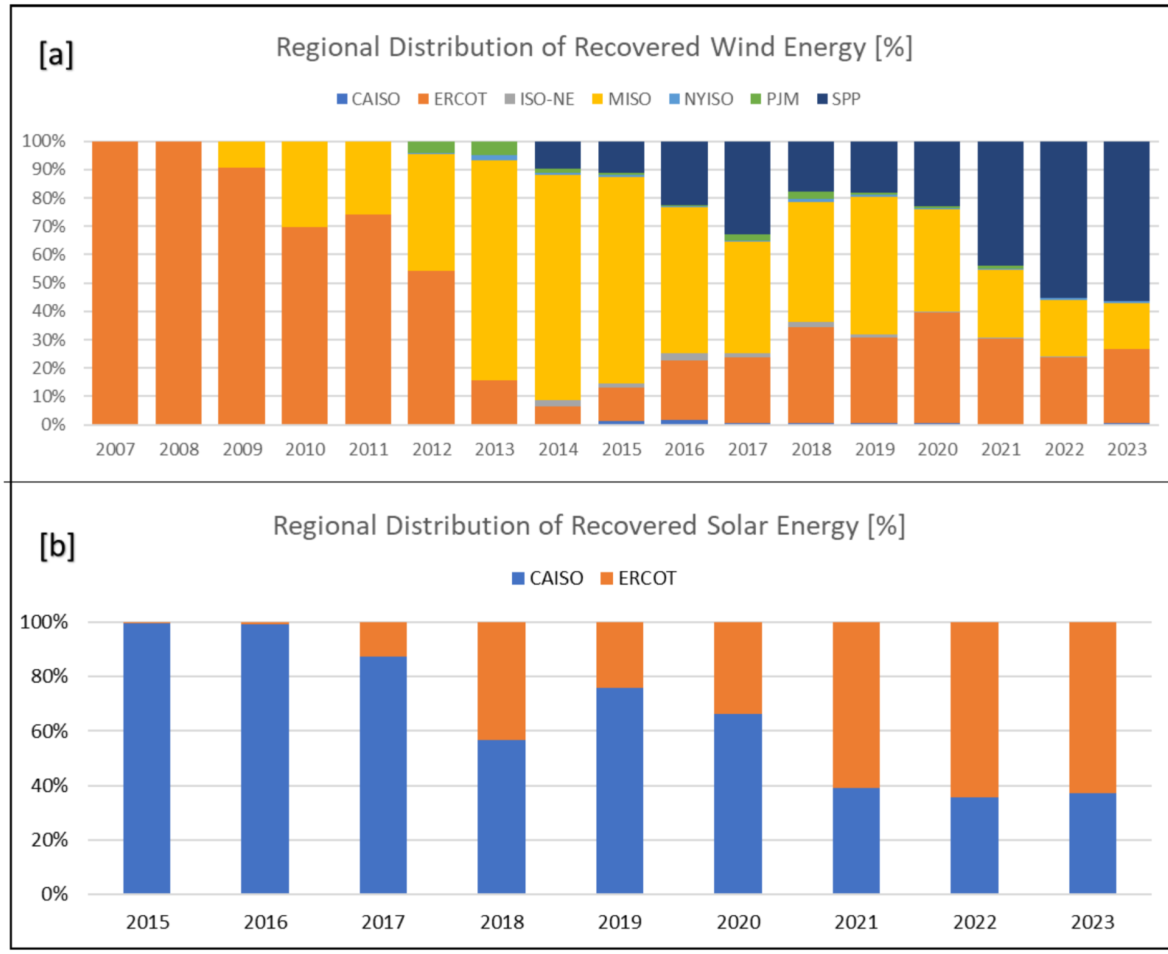

Figure 3—Estimated Average U.S. curtailments from 2007 to 2024: (a) from wind sources; (b) from solar sources. Graphs generated using data from EIA (2024), Wiser et al. (2024), and Seel et al. (2024).

Fig. 4a and Fig. 4b illustrate the recoverable energy over the years from wind and solar sources using subsurface H_2 storage and synthetic geothermal storage technologies, respectively. The technical evaluation shows that subsurface H_2 storage achieved an average total efficiency of approximately 40%, compared to only 14% for synthetic geothermal storage. This efficiency gap is largely attributed to the lower electrical energy requirements associated with subsurface H_2 storage, compared to subsurface geothermal storage.

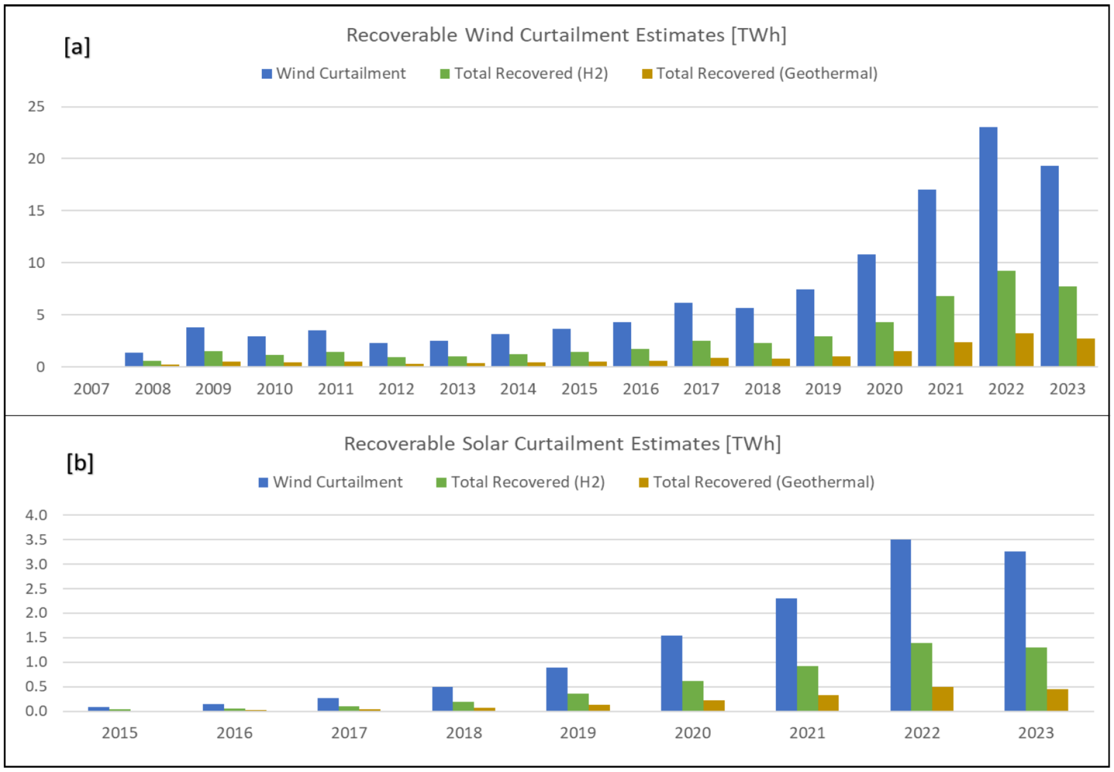

Figure 4—Estimated Average U.S. curtailments from 2007 to 2024: (a) from wind sources; (b) from solar sources. Graphs generated using data from EIA (2024), Wiser et al. (2024), and Seel et al. (2024).

Table 4 provides a summary of the estimated annual average electricity requirements for subsurface H_2 storage across different U.S. regions. Pumping energy to shallow depths ranged from approximately 567 MWh in low-generation areas like NYISO to about 32,000 MWh in high-generation areas like ERCOT. Compression requirements varied from 338 MWh to nearly 19,000 MWh. When storing hydrogen at greater depths, these energy requirements increased—compression requirements rose by approximately 42%, while pumping requirements increased by around 10%. Total annual energy requirements for deep injections ranged from just over 1,000 MWh in ISO-NE and NYISO to nearly 70,000 MWh in ERCOT. These differences reflect regional variations in hydrogen production potential, which is influenced by the availability of curtailed renewable energy. Hydrogen production via both alkaline and PEM electrolyzers yielded similar outputs, confirming strong potential for converting excess wind and solar power into storable and recoverable hydrogen. Among the regions analyzed, ERCOT had the highest average annual recoverable energy at around 44%, followed by MISO (36%), and SPP (17%), while all other regions contributed less than 2%.

Table 4—Estimated annual average electrical energy requirements for subsurface H2 storage at shallow and deep injection by region.

Depth	Electric Requirement	CAISO	ERCOT	ISO-NE	MISO	NYISO	PJM	SPP
Shallow	Compression [MWh]	2,759	19,504	306	15,901	338	375	17,263
	Pumping [MWh]	4,623	32,679	512	26,643	567	629	28,925
	Total [MWh]	**7,382**	**52,183**	**818**	**42,544**	**905**	**1,004**	**46,188**
Deep	Compression [MWh]	4,775	33,756	529	27,520	585	650	29,877
	Pumping [MWh]	5,125	36,223	568	29,532	628	697	32,061
	Total [MWh]	**9,900**	**69,979**	**1,097**	**57,052**	**1,213**	**1,347**	**61,938**

On the other hand, Table 5 presents the estimated annual average electrical energy requirements for synthetic geothermal storage at both shallow and deep depths across the ISO regions. The energy requirements for this technology were significantly higher than those for subsurface H_2 storage, primarily due to the intensive requirements of the condensation and heating processes. For shallow systems using electric boilers, annual energy consumption for these two components alone exceeded 5 million MWh in ERCOT and reached around 4 million MWh in MISO. Even in lower-generation regions such as NYISO and ISO-NE, the requirements remained substantial, surpassing 90,000 MWh and 80,000 MWh, respectively. CSP-based systems followed similar trends, with total shallow-depth requirements ranging from approximately 66,000 MWh in ISO-NE to over 4 million MWh in ERCOT. Deep storage configurations showed comparable patterns, with slightly reduced condensation and heating requirements but increased pumping energy—rising nearly fourfold across all regions. These values highlight the considerable energy intensity of synthetic geothermal systems, especially in high-generation regions such as ERCOT, SPP, and MISO.

Table 5—Estimated annual average electrical energy requirements for synthetic geothermal storage at shallow and deep injection by region.

Depth	Electric Requirement	CAISO	ERCOT	ISO-NE	MISO	NYISO	PJM	SPP
Shallow	Condensation [MWh]	593k	4.1M	66k	3M	73k	81k	3.7M
	Heating [MWh]	770k	5.4M	85k	4M	94k	105k	4.8M
	Pumping [kWh]	3k	23k	364	19k	402	447	21k
	Total (CSP) [MWh]	**593k**	**4.1M**	**66k**	**3M**	**73k**	**81k**	**3.7M**
	Total (Boiler) [MWh]	**770k**	**5.4M**	**85k**	**4M**	**94k**	**105k**	**4.8M**
Deep	Condensation [MWh]	574k	4.0M	64k	3M	70k	78k	3.5M
	Heating [MWh]	745k	5.2M	83k	4M	91k	101k	4.6M
	Pumping [kWh]	13k	90k	1.4k	73k	2k	1.7k	79k
	Total (CSP) [MWh]	**574k**	**4.0M**	**64k**	**3M**	**70k**	**78k**	**3.5M**
	Total (Boiler) [MWh]	**745k**	**5.2M**	**83k**	**4M**	**91k**	**101k**	**4.6M**

Based on Table 6, the annual average hydrogen production and associated water requirements vary significantly across regions and energy sources. ERCOT and SPP show the highest hydrogen production from wind at approximately 33 million kg and 34 million kg, respectively, driven by high levels of renewable curtailment. In contrast, regions like ISO-NE, NYISO, and PJM produced less than 1 million kg each. Hydrogen production from solar was estimated only for CAISO and ERCOT, with 5.1 million kg and 5.8 million kg, respectively. These regions had the largest water requirements, with ERCOT using about 448 million liters in total and CAISO around 63 million liters. Water heating demands, which are relevant for geothermal applications, were notably high in solar-rich areas; CAISO and ERCOT required approximately 1,082 million kg and 7,651 million kg of heated water, respectively. This highlights the impact of regional energy availability and source type (wind vs. solar) on hydrogen production potential. It is also important to know that the hydrogen production reported here assumes no losses due to geochemical reactions or microbial interactions. Indro et al. (2024) reported that hydrogen recovery when potential losses are accounted for could range from 13% to 83% for both sandstone and carbonate reservoirs. Thus, depending on a reservoir's properties, the values reported in Table 6 for hydrogen production could be less.

Table 6—Estimated annual H2 production and water requirements as well as heated water mass by region and source type (wind and solar).

Parameters	CAISO	ERCOT	ISO-NE	MISO	NYISO	PJM	SPP
H2 Production - Wind [Kg]	454k	33M	617k	32M	682k	757k	34M
H2 Production - Solar [Kg]	5.1M	5.8M	-	-	-	-	-
Water Requirement - Wind [L]	5.1M	381M	7.0M	365M	7.7M	8M	396M
Water Requirement - Solar [L]	58M	67M	-	-	-	-	-
Total H2 Produced	**5.5M**	**39M**	**617k**	**32M**	**682k**	**757k**	**34M**
Total Water Requirement	**63M**	**448M**	**7.0M**	**365M**	**7.7M**	**8M**	**396M**
Heated Water - Wind [Kg]	88M	6,509M	119M	6,238M	132M	147M	6,772M
Heated Water - Solar [Kg]	994M	1,142M	-	-	-	-	-
Total Heated Water [Kg]	**1,082 M**	**7,651M**	**119M**	**6,238M**	**132M**	**147M**	**6,772M**

Table 7 presents a comparative summary of the levelized cost of energy (LCOE) for the two storage technologies across the ISO regions. The results revealed significant regional and technological cost differences. Subsurface hydrogen storage shows the highest costs in CAISO and NYISO ($135/MWh) and the lowest in SPP ($86/MWh). In contrast, synthetic geothermal storage using concentrated solar power (CSP) appears more cost-effective, with low LCOE values in CAISO and ERCOT, suggesting that these regions benefit from strong solar availability. Conversely, synthetic geothermal storage powered by electric heating powered by a combined wind and solar energy is the least economical option, with LCOEs ranging from $430/MWh in regions like ERCOT and MISO to $473/MWh in CAISO. These high costs likely stem from the inefficiencies and energy intensity of using electricity from wind and solar combined as a heat source, especially in regions with high electricity prices. Overall, the LCOEs for subsurface hydrogen storage tend to fall just above the $100/MWh range, while synthetic geothermal options—particularly those using CSP—remain below $100/MWh in several regions. These values are broadly consistent with Lazard's (2021) estimates for renewable energy generation: $26–100/MWh for wind, $85–158/MWh for solar PV, and $56–93/MWh for geothermal.

Table 7—Estimated levelized cost by region for subsurface H2 and synthetic geothermal systems over the observed years.

Est. Levelized Cost [$/MWh]	CAISO	ERCOT	ISO-NE	MISO	NYISO	PJM	SPP
Subsurface H2 Storage	135	131	126	118	135	132	86
Synthetic Geothermal Storage (CSP)	98	55	-	-	-	-	-
Synthetic Geothermal Storage (Wind & Solar - Electric Heating)	473	430	461	430	458	455	430

Table 8 presents a regional comparison of average annual energy production and the associated life-cycle CO_2 emissions for various components of the subsurface H_2 storage system. Notably, wind energy dominates ERCOT and SPP, while solar energy contributes significantly to both CAISO and ERCOT. The life-cycle emissions for system components such as solar PV and CSP range reflect emissions from manufacturing, installation, and decommissioning as well. Wind technology exhibits much lower emissions, ranging from 176 tons of CO_2 in CAISO to over 13,000 tons in SPP. Hydrogen processing and pipeline transport introduce additional emissions across all regions. The most striking figures are associated with fuel cell systems: while CAISO shows relatively modest emissions for wind-based fuel cells (~3,246 tons of CO_2), regions like ERCOT and SPP report emissions exceeding 200,000 tons. Overall, the data highlights how both regional energy profiles influence the environmental footprint of H2 systems.

Table 8—Estimated Life-Cycle CO_2 emissions for subsurface H2 storage components by region.

Parameters	Source	CAISO	ERCOT	ISO-NE	MISO	NYISO	PJM	SPP
Avg. Annual Energy Produced	Wind	13k	997k	18k	956k	20k	22k	1M
	Solar	287k	330k	-	-	-	-	-
Life-Cycle Emission [Ton of CO2]	Solar PV Technology	13k	15k	-	-	-	-	-
	CSP Technology	8k	9k	-	-	-	-	-
	Wind Technology	176	12k	239	12k	264	293	13k
	H2 Processing - Wind	191	14k	259	13k	287	318	14k
	H2 Processing - Solar	4.0k	4.6k	-	-	-	-	-
	Pipeline - Wind	41	3k	56	2k	62	69	3k
	Pipeline - Solar	878	1009	-	-	-	-	-
	Fuel Cell - Wind	3k	239k	4.4k	229k	4.8	5k	249k
	Fuel Cell - Solar	69k	79k	-	-	-	-	-

Table 9 summarizes the estimated annual CO_2 emissions for CSP-based geothermal systems and those using electric boilers across different regions. Emissions from geothermal systems powered by wind are minimal in regions like CAISO (0.18 tons) and ISO-NE (0.24 tons), reflecting lower electricity usage. However, emissions increase notably in ERCOT, MISO, and SPP (around 11–12 tons), likely due to the larger-scale energy generation. Solar-powered electric boiler systems show slightly higher emissions (5.87–6.74 tons), consistent with the generally higher life-cycle emissions of solar PV compared to wind. Geothermal systems coupled with CSP also exhibit moderate emissions (3.61–4.15 tons).

Table 9—Estimated CO_2 emissions for the different geothermal systems by region.

Parameters	Source	CAISO	ERCOT	ISO-NE	MISO	NYISO	PJM	SPP
Avg. Annual Energy Produced [MWh]	Wind	12k	906k	16k	869k	1k	20k	943k
	Solar	128k	147k	-	-	-	-	-
System's Emissions [Ton of CO2]	Geothermal - Electric Boiler (Wind)	0.18	11.81	0.24	11.32	0.26	0.29	12.28
	Geothermal Electric Boiler (Solar)	5.87	6.74	-	-	-	-	-
	Geothermal - CSP	3.61	4.15	-	-	-	-	-

Conclusions

This study investigates major U.S. ISO regions' wind and solar generation profiles, the associated curtailment dynamics, and the comparative performance of two long-term energy storage technologies: subsurface H_2 storage and synthetic geothermal storage. The analysis focuses on key metrics, including energy efficiency, CO_2 emissions, and levelized cost.

Between 2007 and 2023, wind and solar energy generation increased substantially across all ISO regions. Wind power growth was most prominent in SPP, ERCOT, and MISO, while solar generation was high in both CAISO and ERCOT. This rapid expansion has led to increased curtailment, particularly in high-generation regions, highlighting the need for efficient and scalable energy storage systems to store excess renewable electricity for future use.

Subsurface H_2 storage demonstrates clear advantages in technical performance, with an average system efficiency of approximately 40%, relatively low life-cycle emissions, and moderate energy requirements for compression and injection. Its effectiveness is particularly pronounced in regions such as ERCOT, MISO, and SPP, where curtailment levels and renewable generation potential are high.

In contrast, synthetic geothermal storage exhibits significantly higher operational energy consumption, especially for condensation and heating using electric boilers. These requirements can translate into greater CO_2 emissions and cost, particularly in regions with high electricity prices. Systems utilizing CSP offer improved cost-competitiveness in solar-rich areas but remain less favorable than H_2 storage in general due to its low system's efficiency.

To conclude, the suitability of long-duration storage technologies is highly region-dependent, influenced by the availability of curtailed renewable energy, expected carbon footprint and electricity prices. Subsurface hydrogen storage, barring losses due to geochemical and microbial reactions, emerges as a more promising solution for maximizing renewable energy utilization and mitigating curtailment in the U.S. Future works will consider losses associated with underground hydrogen storage in porous media to evaluate the technical and economic feasibility.

References

1. Ali, A., Koch, T.W., Volk, T.A., Malmsheimer, R.W., Eisenbies, M.H., Kloster, D., Brown, T.R., Naim, N., and Therasme, O. (2022). The Environmental Life Cycle Assessment of Electricity Production in New York State from Distributed Solar Photovoltaic Systems. *Energies*. 2022; **15**(19): 7278. https://doi.org/10.3390/en15197278
2. Argyrou, M.C., Christodoulides, P., Marouchos, C.C., Kalogirou, S.A., Florides, G.A., and Lazari, L. (2016, June 26). Overview of Energy Storage Technologies and a Short-term Storage Application for Wind Turbines. Paper presented at The 26th International Ocean and Polar Engineering Conference, Rhodes, Greece, June 2016.
3. Aviles, D., Sabri, F., and Hooman, K. (2021). Techno-Economic Analysis of a Hybrid Solar-Geothermal Power Plant Integrated with a Desalination System. *International Journal of Energy Research, Int J Energy Res*, **45**, 12, 0363–907X. https://doi.org/10.1002/er.6948
4. Beckers, K.F., McCabe, K. (2019, February 13). GEOPHIRES v2.0: Updated Geothermal Techno-Economic Simulation Tool. *Geotherm Energy* **7**, 5 (2019). https://doi.org/10.1186/s40517-019-0119-6
5. Bionaz, D., Marocco, P., Ferrero, D., Sundseth, K., and Santarelli, M. (2022). Life Cycle Environmental Analysis of a Hydrogen-Based Energy Storage System for Remote Application., *Energy Reports*, Volume **8**, 2022, Pages 5080–5092, ISSN 2352-4847, https://doi.org/10.1016/j.egyr.2022.03.181
6. Bolinger, M., Seel, J., Kemp, J.M., Warner, C., Katta, A., and Robson, D. (2023, October). Empirical Trends in Development, Technology, Cost, Performance, PPA Pricing, and Value in the United States. Lawrence Berkeley National Laboratory, Utility-Scale Solar, 2023 Edition. https://emp.lbl.gov/sites/default/files/utility_scale_solar_2023_edition_slides.pdf
7. Chen, F., Ma, Z., Nasarabadi, H., Chen, B., Zakaria, M., Mehana, S., and Wijk, J.V. (2023). Capacity Assessment and Cost Analysis of Geologic Storage of Hydrogen: A Case Study in

Intermountain-West Region USA. *International Journal of Hydrogen Energy*, Volume **48**, Issue 24, 2023, Pages 9008–9022, ISSN 0360-3199, https://doi.org/10.1016/j.ijhydene.2022.11.292.

8. EIA. (2024). Power Plant Operations Report: Net Generation by State by Type of Producer by Energy Source. U.S. Energy Information Administration. https://www.eia.gov/electricity/data/state/
9. El-Emam, R.S. and Özcan, H. (2019). Comprehensive Review on the Techno-Economics of Sustainable Large-Scale Clean Hydrogen Production. *Journal of Cleaner Production*, Volume **220**, 2019, Pages 593–609, ISSN 0959-6526, https://doi.org/10.1016/j.jclepro.2019.01.309
10. Giap, V.T., Lee, Y.D., Kim, Y.S., Bui, T., Ahn, K.Y. (2022). New Definition of Levelized Cost of Energy Storage and its Application to Reversible Solid Oxide Fuel-Cell, *Energy*. Volume **239**, Part C, 2022, 122220, ISSN 0360-5442, https://doi.org/10.1016/j.energy.2021.122220.
11. Hoza, M. (2022, May 25). Wind and solar curtailments in ERCOT. *BTU Analytics*. https://btuanalytics.com/power-and-renewables/wind-and-solar-curtailments-in-ercot/
12. Indro, A.P., Sekar, L.K., Matey-Korley, G.V., Ikeokwu, C.C., and Okoroafor, E.R. (2024). A compilation of losses related to hydrogen storage in porous media: Implications for hydrogen recovery and productivity from saline aquifers. *International Journal of Hydrogen Energy*. https://doi.org/10.1016/j.ijhydene.2024.06.365.
13. Kramens, J., Valtere, M., Krigers, G., Kirsanovs, V., Blumberga, D. (2024). Ranking of Independent Small-Scale Electricity Generation Systems. *Clean Technol*. 2024, **6**, 140–151.
14. Khan, M.H.A., Daiyan, R., Han, Z., Hablutzel, M., Haque, N., Amal, R., and MacGill, I. (2021). Designing Optimal Integrated Electricity Supply Configurations for Renewable Hydrogen Generation in Australia. *iScience*, Volume **24**, Issue 6, 2021, 102539, ISSN 2589-0042, https://doi.org/10.1016/j.isci.2021.102539.
15. Lazard. (2021, October 28). Levelized Cost of Energy, Levelized Cost of Storage, and Levelized Cost of Hydrogen 2021. https://www.lazard.com/research-insights/levelized-cost-of-energy-levelized-cost-of-storage-and-levelized-cost-of-hydrogen-2021/
16. Lord, A.S., Kobos, P.H., and Borns, D.J. (2014). Geologic Storage of Hydrogen: Scaling Up to Meet City Transportation Demands. *International Journal of Hydrogen Energy*, Volume **39**, Issue 28, 2014, Pages 15570–15582, ISSN 0360-3199, https://doi.org/10.1016/j.ijhydene.2014.07.121.
17. Okoroafor, E.R., Saltzer, S.D., and Kovscek, A.R. (2022). Toward underground hydrogen storage in porous media: Reservoir engineering insights. *International Journal of Hydrogen Energy*, Volume **47**, Issue 79, 2022, Pages 33781–33802, ISSN 0360-3199, https://doi.org/10.1016/j.ijhydene.2022.07.239.
18. Salmachi, A., Hosseini, T., Shaikh, R.A., Dinovitser, A. and Abbott, D. (2022). Techno-Economic Assessment of Hydrogen Pipe Storage in Decommissioned Wellbores Sourced from Surplus Renewable Electricity. *International Journal of Hydrogen Energy*, Volume **47**, Issue 56, 2022, Pages 23710–23720, ISSN 0360-3199, https://doi.org/10.1016/j.ijhydene.2022.05.160.
19. Seel, J., Kemp, J.M., Cheyette, A., Millstein, D., Gorman, W., Jeong, S., Robson, D., Setiawan, R., and Bolinger, M. (2024). Utility-Scale Solar, 2024 Edition. Lawrence Berkeley National Laboratory. https://emp.lbl.gov/utility-scale-solar/
20. Schaefer, J.M.L. (2022, June). Concept Analysis and Life Cycle Assessment of Large-Scale, Offshore Hydrogen Production, Storage and Distribution Options. University of Bergen Department of Physics and Technology. Thesis. https://hdl.handle.net/11250/3000765
21. Tayyib, D., Elsayed, T., and Okoroafor, E.R. (2025). Techno-economic and life-cycle assessment of subsurface hydrogen and synthetic geothermal storage technologies, *Journal of Energy Storage*, Volume **118**, 2025, 116020, https://doi.org/10.1016/j.est.2025.116020.

22. Thomas, L.K., Tinjum, J.M., and Holcomb, F.H. (2020). Environmental Life Cycle Assessment of a Deep Direct-use Geothermal System in Champaign, Illinois. Stanford University. https://pangea.stanford.edu/ERE/db/GeoConf/papers/SGW/2020/Thomas.pdf
23. Wagner, S.J. (2011, August). Environmental and Economic Implications of Thermal Energy Storage for Concentrated Solar Power Plants. Carnegie Mellon University. Thesis. https://doi.org/10.1184/R1/6716297.v1
24. Wiser, R., Bolinger, M., Hoen, B., Millstein, D., Rand, J., Barbose, G., Darghouth, N., Gorman, W., Jeong, S., O'Shaughnessy, E., and Paulos, B. (2023). Land-Based Wind Market Report: 2023 Edition. U.S. Department of Energy, Office of Energy Efficiency & Renewable Energy. https://emp.lbl.gov/sites/default/files/2023_lbwmr_final.pdf
25. Wiser, R., Millstein, D., Hoen, B., Bolinger, M., Gorman, W., Rand, J., Barbose, G., Cheyette, Darghouth, Jeong, S., Kemp, J., O'Shaughnessy, E., Paulos, B., and Seel, J. (2024). Land-Based Wind Market Report 2024 Edition. Lawrence Berkeley National Laboratory. https://emp.lbl.gov/wind-technologies-market-report